

A theoretical study of the principles regulating the specificity for Al(III) against Mg(II) in protein cavities

Elixabete Rezabal ^{*}, Jose M. Mercero, Xabier Lopez, Jesus M. Ugalde

Kimika Fakultatea, Euskal Herriko Unibertsitatea and Donostia International Physics Center (DIPC), PK 1072, 20080 Donostia, Euskadi, Spain

Received 16 April 2007; received in revised form 25 May 2007; accepted 5 June 2007

Available online 12 June 2007

Abstract

Several toxic effects arise from Al's presence in living systems, one of them being the alteration of the natural role of enzymes and non-enzyme proteins. Al(III) is capable of entering protein active sites that in normal conditions should be occupied by other metals. Even if Mg(II) is an ubiquitous metal in biological systems, the interference of aluminium in magnesium metabolism is not clear yet. In this work, a systematic study of the affinity of Al(III) for different protein binding sites is presented, with special attention on structural parameters, the role of the charge and the presence of different ligands in the protein cavity. The specificity of the binding site for Al(III) against Mg(II) has been studied, and also the thermodynamical propensity of a Mg(II)/Al(III) exchange. Quantum mechanical methods that proved to be reliable in previous works have been used, namely, the density functional theory (DFT) and polarizable continuum model (PCM).

© 2007 Elsevier Inc. All rights reserved.

Keywords: Metal specificity; Aluminum; Magnesium; Metal exchange; DFT; Dielectric effects

1. Introduction

The key role of the metal ions in biochemistry, as regulators of the activity of a myriad of enzyme and non-enzyme proteins, attracts the interest of an increasing number of studies each year. Providing accurate metal-site structures in proteins is crucial for several studies regarding the design of drugs [1,2], metal toxicity, specificity of proteins or development of new techniques [3]. Lately, the importance of the combination of experimental and computational works has been addressed [4,5], specially when studying the structure and thermodynamics of the metal binding to a protein.

This study recalls the toxicity of the Al(III) in biological systems. Al(III) is a non essential element that has been included in the living systems mainly due to human intervention [6]. It interacts with proteins, altering their conformation and enhancing toxic processes and also interfering

with the metabolism of essential metals, like Mg(II), Ca(II) and Fe(III) [7,8]. The interference of Al(III) in Mg(II) metabolism is by far the least studied, even though it has been well documented [7]. Consequently, its molecular basis remains unknown. Mg is one of the most abundant elements in biological environments, and it has a key role as a cofactor of several kinds of proteins in living systems. The activity of several of these enzymes has been found to be altered by Al(III) [7]. Al(III) has been seen to enter binding sites that in healthy systems should be filled up by other metals, either by occupying a site which is usually empty or by substituting the metal. Some authors even point out that some altered proteins may present an increased affinity for Al(III) and a decreased affinity for Mg(II), what disrupts the protein's function [9] and leads to neurological problems. Even this feature is still highly hypothetical, due to the scarcity of reliable experimental data available, the physicochemical similarities of both metals make this possibility non negligible [10].

Our scope is the systematic study of the affinity of Al(III) for different protein binding sites and the conditions

^{*} Corresponding author. Tel.: +34943015341.

E-mail address: pobrease@sc.ehu.es (E. Rezabal).

in which a Mg(II)/Al(III) metal exchange could take place. This substitution has already been seen to be able of blocking the D-xylose isomerase [11]. In previous works, we studied the affinity of both metals for the possible ligands present in the binding sites of the proteins [12], and the role of the carboxylate ligand [13]. In our last work, the capability of Al(III) to substitute Mg(II) in its most frequent natural binding sites was studied [14].

Related to that work, the present paper reports on a systematic study of the affinity of Al(III) and Mg(II) for three ligand binding sites (most frequent binding sites for Mg(II) [15]), focusing on the evaluation of the structural and energetical parameters of their binding to the proteins. The role of the carboxylate binding mode and the neutral ligands will be carefully studied.

2. Molecular models for the active site

In protein active sites, the metal is bound to the side chains of the aminoacids. This immediate environment will dictate the biological activity of the metalloprotein, since the first shell ligands play crucial roles in contributing to the metal complex stability and in determining the selectivity of the binding site [4,5]. Considering that the geometry of metal sites in proteins is very close to that obtained in vacuum, we will model the active site with molecular complexes formed by the metal and its first coordination shell, which will be composed by three bio ligands (side chains of the aminoacids most frequently found in proteins) and water molecules.

Five different ligands, which have shown the highest affinity for both Mg(II) and Al(III) in previous works, [12] have been considered: negatively charged acetate (to simulate aspartate and glutamate) and methylthiolate (for deprotonated cysteine), and the neutral methylimidazole (for Histidine), acetamide (for asparagine, glutamine and the backbone carbonyl groups) and methanol (for serine and threonine). Protein Data Bank (PDB) surveys found in the literature [15] and the results obtained by our group in previous works [12,13] reveal the low probability of finding binding sites not containing at least one acetate. Hence, in the present work, we will always include at least one acetate in the binding site, bound in a monodentate or a bidentate way. Different combinations of the remaining two ligands will be studied, forming binding sites with charges -3 , -2 and -1 .

In an aqueous environment, the structure and number of water ligands in the various hydration shells of solvated metals are a central issue [16], in particular for the calculation of their dynamical properties [17]. Structural properties, on the other hand, are less sensible to the hydration shell structure. So that, consideration of only the first hydration shell yields often reliable results [18], which for the specific cases of Mg(II) [19] and Al(III) [20], has been established to contain six water ligands, in both cases. In protein environments, X-ray studies suggest that the metals are usually coordinated to water molecules even in buried

sites of the protein; consequently, we will add up to the metal the necessary water molecules as to fill up the first solvation shell of the metals, which are both found octahedrally coordinated in biological environments.

The protein surrounding the metal cavity induces a long-range polarization on the active site [4]. The polarizable continuum model (PCM) gives us the possibility of simulating the polarization in different parts of the protein. The binding site may be located in a more or less solvent accessible area of the protein. The more buried the site is, the less polarizable it will be. We will consider four different situations: the first will not take into account any environmental effect, and thus, the intrinsic propensity of the reactions will be studied (gas phase data). A buried site of the protein will have the polarizability dictated by the protein, which is usually considered to have dielectric constant $\epsilon = 4$. A more solvent exposed area of the protein will present a higher dielectric constant, ($\epsilon = 20$). Totally solvent exposed binding sites have water's polarizability, thus, their dielectric would be around 80.

3. Computational methods

All calculations were carried out with the GAUSS-IAN03 [21] package. It has already been established that density functional methods give reliable results in most chemical systems [22,23]. Density Functional Theory (DFT)/Hartree–Fock (HF) hybrid methods are particularly appealing for the present investigation since they correct the pure DFT overestimation of the bond dissociation energies [24] as was validated by Johnson et al. [25]. We have chosen the B3LYP hybrid DFT approximate functional [26] for this work, which consists of the B3 exchange functional [27], the LYP correlation functional [28] and a 20% of exact exchange. It has already been well established that this density functional implementation gives excellent results for most chemical systems [29] including organometallic compounds.

The standard all electron 6-31+G(d,p) basis set was used for the metal cations, while the compact effective core potentials and shared-exponent basis set of Stevens, Basch, Krauss and Jasien (SBKJ) [30] including a polarization and a diffuse function was used for atoms of the ligands. Gresh et al. [31–33] found that this pseudopotentials/all-electron basis set combination for the ligand and the metal cation, respectively, represent a very well balanced compromise between accuracy and computational efficiency. This method has been widely used in our group and has shown to be adequate for this type of calculations [1,12,34–37], as the results are found to compare satisfactorily with the scarcely available experimental data and with the theoretical data obtained by other groups with more flexible basis sets on similar systems [12]. This basis set will hereafter be referred as SBKJ/*+.

Frequencies, thermal corrections (E_{TRV}) and entropies were calculated at this level of theory and the corresponding zero-point vibrational energy (ZPVE) corrections made

to the total energy, using standard statistical mechanical methods [38] at 298 K. The enthalpy and free energy changes of the complexation $M + nL \rightarrow M(L)_n$ were evaluated at 298 K as follows:

$$\Delta H = \Delta E_{\text{elec}} + \Delta E_{\text{TRV}} + \Delta \text{ZPVE} + n\Delta \text{PV} \quad (1)$$

$$\Delta G = \Delta H - T\Delta S_{\text{TRV}} \quad (2)$$

The gas phase energies were recalculated on the B3LYP/SBK*+ geometries at the B3LYP/6-311++G(2df,2p) level of theory, and corrected with the ZPVE and thermodynamical data calculated at the B3LYP/SBK*+ level of theory.

Aqueous phase optimizations were carried out using the self-consistent reaction field method (SCRF), by the integral equation formulation polarizable continuum model (IEFPCM) [39–41] procedure at the B3LYP(IEFPCM)/SBKJ/*+ level of theory. Solvation free energies were also recalculated on the B3LYP(IEFPCM)/SBKJ/*+ geometries at the B3LYP(IEFPCM)/6-311++G(2df,2p) level of theory.

Continuum models of solvation constitute one efficient way to include condensed-phase effects into quantum mechanical calculations, by means of using a self-consistent reaction field (SCRF) approach for the electrostatic component. However, this approach does not include the non-electrostatic component of solvation, which in some cases might be non-negligible [42]. Additionally, local effects arising from the inner hydration shells around solvated ions are treated only partially. Hence, critical consideration of the number of explicit solvent molecules that are necessary to include, in order to account reliably for the effects of hydration shell structure, is required when using PCM. Ca(II) seems to be one of these cases [12].

However, earlier calculations have shown that for the specific cases of Al(III) and Mg(II), and also for Zn(II), PCM does yield reliable results, as shown by our earlier validation calculations on the relative free energies of these systems [12].

Gas phase corrections were employed to estimate the aqueous phase thermodynamic properties as found in the literature [43,12], since vibrational frequencies calculation in liquid phase was computationally very demanding (and not very accurate) and corrections were expected to be very similar [13].

Solvation energy of each structure in dielectrics $\epsilon = 4$ and $\epsilon = 20$ were calculated at the HF(PCM)/6-31+G**/B3LYP/SBKJ/*+ level of theory.

The validation of the method and comparison with experimental data for similar systems was carried out in previous works with satisfactory results [37,12,13].

4. Results

4.1. Evaluation of the structural parameters

Several isomers were studied for each ligand combination, and the most stable conformations are shown in

Fig. 1. Overall, very similar structures were obtained for both metals.

Binding sites with charge -3 can be formed by either three acetates, two acetates and one methylthiolate or one acetate and two methylthiolates. Both monodentate and bidentate ways of binding were considered for the acetate.

Thus, we have four different ligand combinations when considering three acetates in the binding site (3B, M-2B, 2M-B and 3M, see Fig. 1 for the complex labeling). Several isomers were found for each of them, up to 16 kcal/mol less stable than the absolute minimum in the case of Al(III) and 9 kcal/mol for Mg(II). The highest stability was achieved with an hydrogen bond between the free oxygen of the monodentate acetate and the hydrogen of a water molecule of the first hydration shell of the metal. This motif is very commonly found in metalloprotein crystal structures [44].

Only two complexes formed by two acetates and one methylthiolate were found (2M-T and MB-T). As a consequence of the steric hindrance arising from S atom's big size, it was not possible to characterize any complex with two bidentate acetates and one methylthiolate, for any of the studied metals. For the same reason, the binding site presenting one monodentate acetate, a bidentate acetate and a methylthiolate ligand did not form any kind of stable hexacoordinated structure with the Mg(II). Al(III) can accommodate the bidentate acetate and the sulfur atom in the first hydration shell, and due to its higher charge as compared to Mg(II) presents a higher affinity for the binding site with charge -3 . Therefore, several MB-T structures were found for Al(III) falling in a range of 13 kcal/mol. The most stable isomer for 2M-T complex presents both acetates in axial–axial orientation with respect to each other; other orientations presented 14 kcal/mol less stable formation energies for Al(III) and 6 kcal/mol for Mg(II).

Complexes consisting of two methylthiolates and one acetate showed a smaller range of energies (2 kcal/mol for Al(III) and 1.8 kcal/mol for Mg(II)), and in both cases the preferred orientation of the ligands was equatorial–axial (M-2T and B-2T).

Nine different complexes were optimized for dianionic cavities (binding sites with one neutral ligand and two acetates). The absolute minima found for each coordination fashion of the acetate (2B-X, 2M-X and MB-X) was very similar for the three neutral ligands studied, as can be seen in Fig. 1. Equatorial–axial arrangement is favoured in 2M-X and MB-X complexes, instead, in 2B structures, the neutral ligand is preferred to be in axial orientation as respect to the acetate.

Finally, when two neutral ligands are found in the binding site, monoanionic cavities are formed. The most stable configuration presents both neutral ligands in axial–axial (aa) position with respect to one another, when the acetate is monodentate, and all the ligands in axial–equatorial position when it is bidentate. The found isomers fall in a

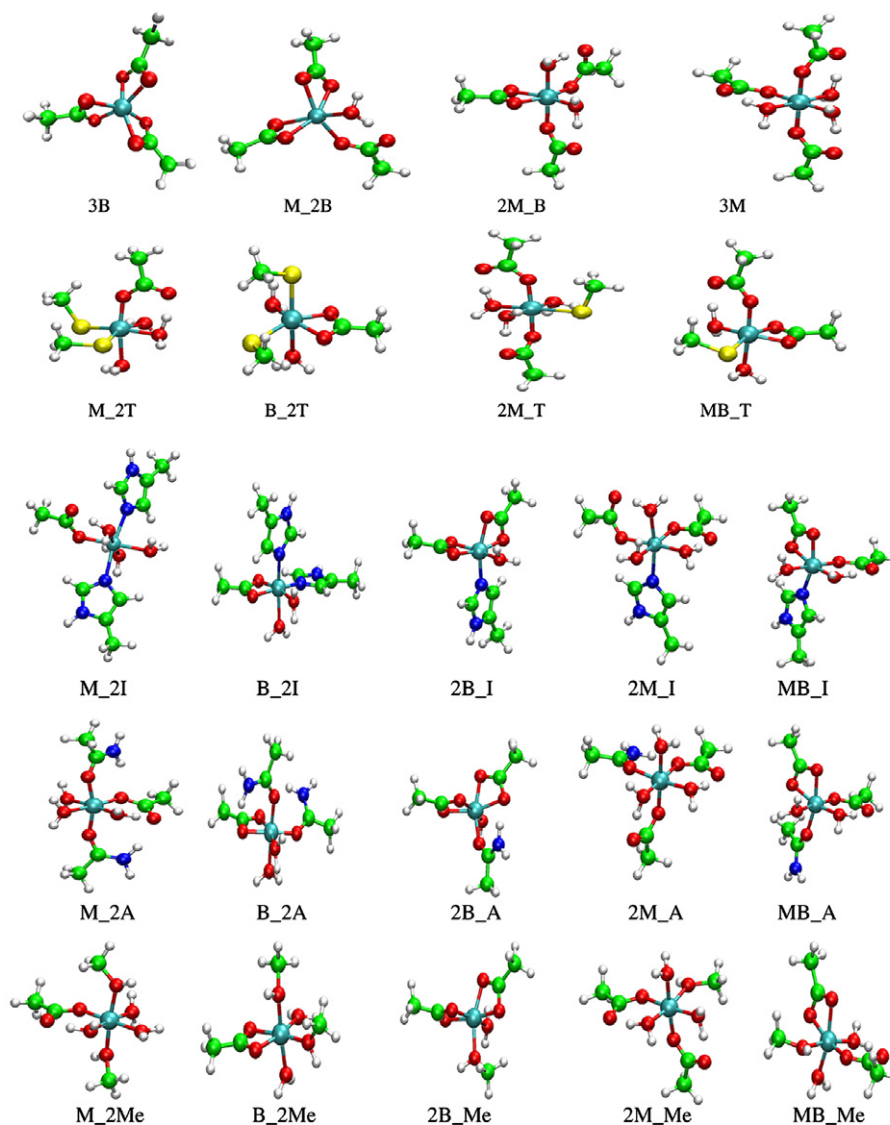


Fig. 1. Complexes studied. Figures correspond to the aluminum aqueous phase geometries where magnesium complexes are analogous either in aqueous and gas phase. **M** corresponds to a monodentate acetate, **B** is a bidentate acetate, **T** stands for the methanethiolate, **I** for the methylated imidazole, **A** for the acetamide and **Me** for the methanol.

range of 1.4 kcal/mol for Al(III) and 3.2 kcal/mol for Mg(II).

Related to the metal–ligand bond lengths, overall, three tendencies are observed in all the complexes, regardless of the ligands considered: first, Mg(II)–ligand bond lengths are around 0.2 Å larger than the Al(III)–ligand ones (see Fig. 2), due to the weaker electrostatic interaction between the lower-charged Mg(II) and the ligands as compared with the Al(III), and also to the bigger ionic radius of the former (0.72 Å vs. 0.54 Å). Second, no remarkable variation was seen between gas phase and solution bond distances, as expected [43]. Anyway, a subtle difference is noted, which increases when increasing the charge of the complex, namely, that bonds in solution are equal or slightly larger for negative ligands but not for the neutral ligands, in this case, bond lengths decrease in solution as compared to gas

phase. And last, all bond lengths become smaller as the charge of the cavity increases, even this variation is very subtle.

Concerning the acetates, bidentate acetate presents a metal–O_{acetate} bond larger than the monodentate acetates, for both metals (2.14 Å vs. 2.06 Å for Mg(II) and 1.95 Å vs. 1.88 Å for Al(III)) (see Fig. 2). This feature is widely known, and was also found in previous works [13]. When the acetate binds monodentately, more charge is transferred to the metal along each bond, and the metal–oxygen distance becomes shorter. An exception: binding sites containing a neutral ligand, thus, dianionic and monoanionic cavities, presented in gas phase spontaneous proton transfers from a water molecule of the hydration shell to the free oxygen of the monodentate acetates, when bound to Al(III). This feature has also been observed in previous

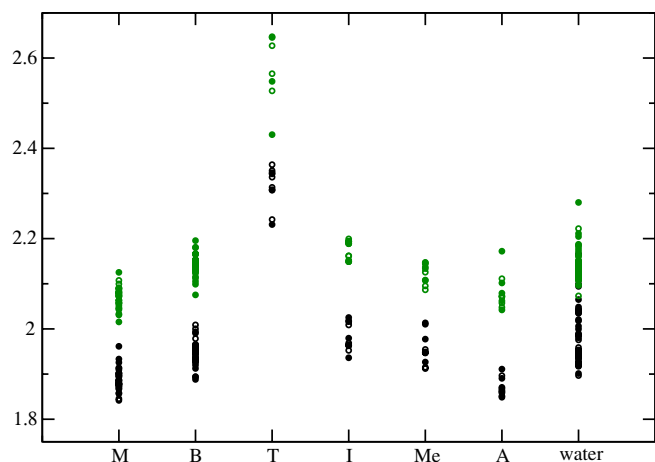


Fig. 2. Gas phase (filled symbols) and aqueous phase (empty symbols) metal–ligand bond lengths (in Å) of the complexes described along the text. The ligands are depicted in the X axis, green symbols stand for Mg(II) complexes and the black ones for Al(III).

works [12,13], as a consequence of the strong acidity of the water molecules, caused by the high charge in Al(III). The proton transfer elongates the Al(III)–O_{acetate} distance, and the difference with the Al(III)–O_{acetate} distance in bidentate acetates is reduced to about 0.02 Å (see [supplementary material](#)). The larger charge transfer to Al(III) in binding sites with charge –3 prevents this proton transfer to occur [44].

Tunega et al. [43] studied Al(III)–acetate complexes, and reported one structure with three bidentate acetates. The Al(III)–carboxylate oxygen distance they reported were slightly longer than ours, 1.966 Å vs. 1.940 Å in gas phase and 1.958 Å vs. 1.937 Å in aqueous phase, but are in general in good agreement considering the different computational level used.

The other negatively charged ligand that was considered, methylthiolate, presents bond lengths 0.5 Å larger than the rest of the ligands, due to the bigger radius of the S atom as compared to the others. The lower affinity of Mg(II) for methylthiolate ligand, as compared to Al(III) can also be seen in the long bonds formed by Mg(II) in the structures containing methylthiolates, where in some cases the ligands are almost taken away from the first hydration shell (see Fig. 2).

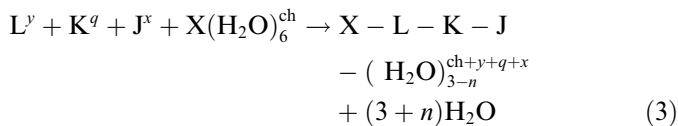
The neutral ligands presented little geometrical variation in all the different complexes studied, as can be seen in Fig. 2. Al(III) presented bond lengths of 1.98, 1.96 and 1.87 Å for methylimidazol, methanol and formamide, respectively, and in Mg(II) complexes the distances were 2.18, 2.13 and 2.09 Å, in the same order.

Metal–O_{water} distances, in gas phase, fall in a range of 2.03–2.01 Å for Al(III) in gas phase and 2.23–2.14 Å for Mg(II). The largest bonds are found for binding sites with charge –3 and the shortest in monoanionic cavities. When proton transfer occurs, the bond is shortened around 0.2 Å, since the metal is now bound to a hydroxile instead of a water molecule (see [supplementary material](#)). In solu-

tion, the bond lengths decrease about 0.06 Å for Al(III) and 0.03 Å for Mg(II), as compared to gas phase.

4.2. Metal binding affinity

The metal binding affinity will be evaluated by calculating the energy of the following reaction:



where ch, y, q and x are the charges of the metal cation and the L, K and J ligands, respectively. n corresponds to the number of acetates bound bidentately, and X stands either for the Al(III) or the Mg(II) cation.

The reaction defines the metal binding affinity as the water/ligand substitution from the first hydration shell of the metal, where all the exchanges occur simultaneously.

The formation enthalpies obtained for each ligand combination and protein environment are depicted in Fig. 3. This data, together with the free energies, are available as [supplementary material](#).

Overall, both metals are very prone to binding the active site composed by three ligands, in all the range of dielectrics inside the protein.

The intrinsic propensity of the thermodynamic reaction (blue symbols) is very favourable. The affinity goes up to –670 kcal/mol for Al(III) and –360 kcal/mol for Mg(II).

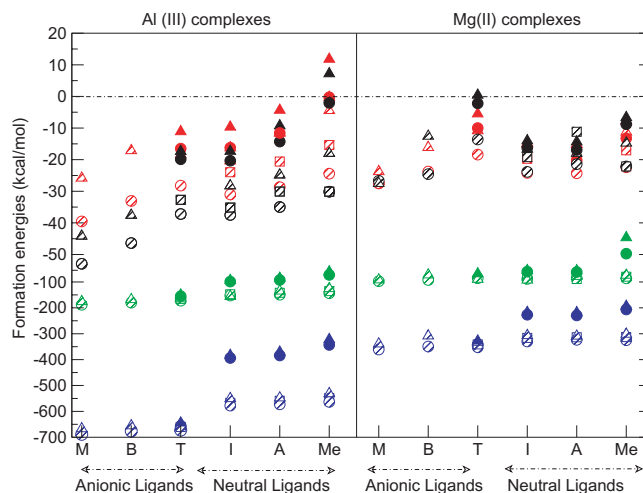


Fig. 3. Formation reaction enthalpies of the complexes in kcal/mol. Filled symbols stand for the complexes containing one acetate plus two ligands defined in the X-axis. Stripped symbols stand for complexes formed by two acetates plus one ligand defined in the X-axis. Triangles stand for bidentate acetates and circles for monodentate acetates. The square stripped symbols denote structures with one monodentate and one bidentate acetates. Note that the energy scale changes at –50 kcal/mol. Blue symbols stand for the gas phase data, green ones for the $\epsilon = 4$, the black for $\epsilon = 20$ and the red ones for aqueous phase enthalpies.

In this case the reaction is driven primarily by favorable charge–charge interactions between the positively charged metal complex and the negatively charged active site.

The charge of the active site (−3, −2 or −1) regulates this affinity, which increases with the charge (in absolute value). This is particularly remarkable in the case of Al(III), as the formation enthalpies obtained are −700, −570 and −400 kcal/mol, respectively. Mg(II), on the other hand, has similar affinity for −3 and −2 charged ligand combinations (−380 and −350 kcal/mol, respectively). The interaction with the three negatively charged ligands is still more favourable than with two negatively charged ligands, but, the energy gain is smaller, since, each metal–ligand bond is weaker in the former, due to the strong charge transfer to Mg(II). The affinity for −1 charged ligand combinations falls around −230 kcal/mol.

These differences are smothered when environmental effects are included. It was already seen in previous works [13] and in the literature [45], that increasing the solvent exposure of the metal binding site attenuates the energy gain of the water/ligand exchange because the desolvation penalty of the charged reactants exceeds the solvation free energy gains of the less charged products. Consequently, the reaction energy decreases in absolute value as more solvent accessible is the area studied.

The lower the net charge in the binding site, the desolvation penalty will be less important, since the charge difference between reactants and products is smaller in this case. In resume, we have two opposed driving forces. On one side, the favourable electrostatic interactions between positively charged metal and the binding site favours the most negatively charged ligand combinations. Instead, they are the most affected by the desolvation penalty. Due to the balance between these two forces, the affinity difference arising from the binding site charge is less pronounced as the solvent exposure increases. Thus, even in gas phase the presence of three negatively charged ligands highly enhances the metal/ligand exchange, in the most solvent exposed areas of the protein, the affinity is similar regardless of the ligand combination.

Although charge is the dominant factor, the nature of the ligands chosen also affects the formation energy of the complexes. Among all the ligands studied, the acetate is the only one which has two binding atoms, thus, it can bind the metal cation via one of the carboxylate oxygen atoms (monodentate mode) or both oxygen atoms (bidentate mode). The mode of carboxylate binding is known to play an important role in recognition of the native metal cofactor and thus, the function of metalloproteins [46].

In gas phase, both metals present the same affinity order for the three acetate complexes. The results show a preference of the metals for monodentate acetates, where the free oxygen of the monodentate acetate is stabilized by the interactions with the water molecules of the first shell [46]. The most favorable formation is that corresponding to the 3M complex, followed by 2MB, M2B and 3B. The

energy gap between complexes is around 10 kcal/mol for both metals, except the 3B complex for Mg(II), which is 30 kcal/mol less stable than the M2B one.

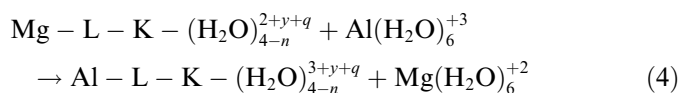
Mg's charge is remarkably reduced by the attachment of three acetates, two of them being bidentate, and thus, binding the remaining free oxygen destabilizes further the complex. This does not happen with Al(III) since it can accept more charge than Mg(II), and the gap between 3B and M2B complexes remains similar in this case. The difference between the complexes decreases as we increase the solvent exposure. The preference of monodentate acetates over bidentates is also seen in dianionic and monoanionic cavities.

In general, when a methylthiolate ligand is included in the binding site, the behaviour is similar to the rest of −3 charge binding sites, even though the formation energies are not so favoured as for three acetate binding sites. The larger size of the S atom results in steric constraints that elongate the bonds, making the interactions weaker. This is reflected specially in Mg(II) complexes, due to the metals low charge and bigger size, where in the most solvent accessible areas of the protein the formation energies are lower than for any other complexes, and the bond lengths increase. This agrees with the fact that in nature, sulfur atoms are seldom found in Mg(II) binding sites.

Between the neutral ligands, as expected, the preferred ligand is methylimidazol, and the less favoured one is the methanol [13]. The latter presented specially low affinity in monoanionic cavities, and in the most solvent exposed areas of the protein ($\epsilon = 20, 78$) its formation was thermoneutral or even unfavourable for Al(III) complexes.

4.3. Metal exchange reaction

The metal exchange reaction will be defined as follows:



The energy balance of this reaction, the exchange energy, will indicate the likelihood for the hydrated Al(III) to substitute a Mg(II) already attached to a binding site in a protein.

In this case we did not consider the methylthiolate ligands, since they are very rarely found in natural Mg(II) binding sites, along with the difficulties experimented when characterizing Mg(II) complexes with methylthiolate ligands (see previous section).

The exchange enthalpies obtained are depicted in Fig. 4 and all the data are presented in detail in the [supplementary material](#). Notice that this data for binding sites consisting of two monodentate acetates plus a ligand was discussed more extensively in a previous work [14], and is presented here (stripped circles) for comparison with the new data.

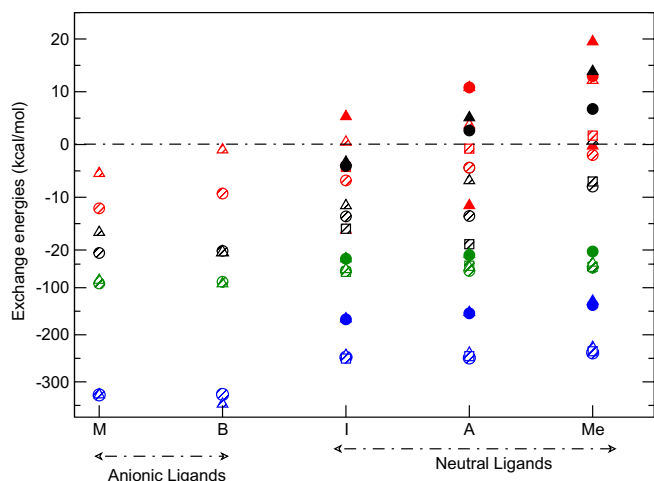


Fig. 4. Exchange reaction enthalpies of the complexes in kcal/mol. Filled symbols stand for the complexes containing one acetate plus two ligands defined in the X-axis. Stripped symbols stand for complexes formed by two acetates plus one ligand defined in the X-axis. Triangles stand for bidentate acetates and circles for monodentate acetates. The square stripped symbols denote structures with one monodentate and one bidentate acetates. Note that the energy scale changes at -20 kcal/mol. Blue symbols stand for the gas phase data, green ones for the $\epsilon = 4$, the black for $\epsilon = 20$ and the red ones for aqueous phase enthalpies.

In general, the exchange reaction is energetically very probable to occur. In gas phase, it can be as favourable as -340 kcal/mol, for the binding sites with three negatively charged ligands. Again, the charge of the cavity plays a major role in the thermodynamics of the reaction. The charge difference between both metals makes the exchange much more favourable the more negative the active site of the proteins.

As solvation effects are included, the desolvation penalty for Al(III) complexes is higher than for Mg(II) ones, because the charge difference between reactants and products is more remarked for the former. So, as the polarization of the site increases, the capacity of aluminium to substitute magnesium decreases, and at some extent, becomes unfavourable. If we have a monoanionic cavity (filled symbols) in relatively solvent-exposed area, the exchange will not be favorable. Dianionic cavities also disfavour the exchange in totally solvent exposed area (red symbols) when two bidentate acetates are present in the complex for acetamide and methanol.

In gas phase, the exchange energy in triply charged binding sites is also most favoured when three bidentate acetates are considered. Charge transfer is highest in these conformation, which favours the attachment of trivalent cations [3]. In the most solvent exposed areas, the preferred conformation for exchange is 3M because of the polarization effect.

When one methylimidazole is present in our active site, the most likely conformation to undergo metal exchange is when one of the acetates is monodentate and the other bidentate. In the rest of the structures containing one or

two neutral ligands the exchange is most favoured when the acetates are monodentately bound.

5. Discussion

Previous extensive surveys of the protein data bank have concluded that most common natural occurring Mg(II) protein binding sites are formed by three ligands, at least one of them being an acetate [15]. Consequently, in this paper we have studied the affinity of both metals, Mg(II) and Al(III), for these active sites and also their selectivity towards the metal. The negative charge of the cavity has been varied by adding negatively charged ligands, since it has been seen that in biological systems the net charge of the binding pocket contributes to the selective binding of higher charged metals against much higher concentration of less charged metals [47].

Both metals present favourable binding affinities, regardless of the ligand combination, in all the protein environments studied. As an exception, monoanionic cavities containing two methanol ligands gave unfavourable binding energies towards Al(III) in the most solvent exposed areas of the proteins, due to the high polarizability of the site and the low affinity for methanol ligands.

Carboxylate has been seen to be the preferred negatively charged ligand, and the highest affinity is achieved in binding sites containing three acetates (net charge -3), specially Al(III), which has been shown to be a better charge acceptor than Mg(II).

Preference for monodentate binding mode has been detected, in accordance with the literature, where it has been found that from all the Mg(II) binding sites with first shell acetate ligands, only one Mg(II) active site was found to contain bidentately bound carboxylates [48]. Additionally, the presence of a monodentate acetate in the binding helps the Mg/Al substitution. The more acetates in the binding site, the more likely the exchange to occur, since Al(III) can accept more carboxylates in its first coordination shell [46].

As compared to previous works [13], the addition of a neutral ligand in an monoanionic or dianionic cavity (consisting of one acetate and one neutral ligand or two acetates, respectively) promotes the affinity of both metals, becoming favourable to bind even in the most solvent exposed areas. This addition also favors the substitution in dianionic cavities, which was unfavourable in the most solvent exposed areas of the protein, when consisting of only two acetates. Nevertheless, the Mg/Al exchange possibility remains unfavourable in the most solvent exposed areas for monoanionic cavities.

Finally, the shorter bond lengths of Al(III) as compared to Mg(II) have been confirmed for all the complexes, due to the higher charge and smaller size of Al(III). This may be the cause of the low exchange rate of ligands in sites served by Al(III) due to its enhance rigidity. The latter is claimed to be the reason for the blockage of Al(III) bound proteins [11].

6. Abbreviations

PDB	Protein Data Bank
M	monodentate acetate
B	bidentate acetate
T	methylthiolate
I	methylimidazol
A	acetamide
Me	methanol
DFT	density functional theory
HF	Hartree–Fock
B3	Becke proposed hybrid
LYP	Lee–Yang–Parr
SBKJ	Stevens, Barsch, Krauss and Jasien
ZPVE	zero-point vibrational energy
E_{TRV}	thermal corrections
SCRf	self-consistent reaction field
IEFPCM	integral-equation-formalism polarizable continuum model

Acknowledgement

This research was funded by Euskal Herriko Unibertsitatea (the University of the Basque Country), Gipuzkoako Foru Aldundia (the Provincial Government of Guipuzkoa), and Eusko Jaurlaritza (the Basque Government). The SGI/IZO-SGIker UPV/EHU (supported by the National Program for the Promotion of Human Resources within the National Plan of Scientific Research, Development and Innovation – Fondo Social Europeo and MCyT) is gratefully acknowledged for assistance and generous allocation of computational resources.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jinorgbio.2007.06.010](https://doi.org/10.1016/j.jinorgbio.2007.06.010).

References

- [1] E. San Sebastian, J.M. Mercero, A.D. Roland, H. Stote, F.P. Cossio, X. Lopez, *J. Am. Chem. Soc.* 128 (2006) 3554–3563.
- [2] R. Ge, H. Sun, *Acc. Chem. Res.* 40 (2007) 267–274.
- [3] T. Dudev, L. Chang, C. Lim, *J. Am. Chem. Soc.* 127 (2005) 4091–4103.
- [4] F. Himo, *Theor. Chem. Acc.* 116 (2006) 232–240.
- [5] U. Ryde, *Dalton Trans.* (2007) 607–625.
- [6] C. Exley, *J. Inorg. Biochem.* 97 (2003) 1–7.
- [7] T. Kiss, M. Hollosi, in: C. Exley (Ed.), *Aluminium and Alzheimer's Disease*, Elsevier, Amsterdam, 2001.
- [8] T. Kiss, K. Gajda-Schranz, P.F. Zatta, in: A. Sigel, H. Sigel, R. Sigel (Eds.), *Neurodegenerative Diseases and Metal Ions*, Wiley, England, 2006.
- [9] J.L. Glick, *J. Theor. Biol.* 148 (1991) 283–286.
- [10] R.B. Martin, *Aluminum in Chemistry Biology and Medicine*, Raven Press, New York, 1991.
- [11] T. Gercei, Z. Bocskei, E. Szabo, B. Asboth, G. Naray-Szabo, *Int. J. Biol. Macro.* 25 (1999) 329–336.
- [12] J.M. Mercero, J.M. Matxain, E. Rezabal, X. Lopez, J.M. Ugalde, *Int. J. Quantum Chem.* 98 (2004) 409–424.
- [13] E. Rezabal, J.M. Mercero, X. Lopez, J.M. Ugalde, *J. Inorg. Biochem.* 100 (2006) 374–384.
- [14] E. Rezabal, J.M. Mercero, X. Lopez, J.M. Ugalde, *Chem. Phys. Chem.*, accepted for publication.
- [15] T. Dudev, J.A. Cowan, C. Lim, *J. Am. Chem. Soc.* 121 (1999) 7665–7673.
- [16] H. Ohtaki, T. Radnai, *Chem. Rev.* 93 (1993) 1157–1204.
- [17] T.S. Hofer, B.R. Randolph, B.M. Rode, *Chem. Phys. Lett.* 422 (2006) 492–495.
- [18] T.S. Hofer, B.R. Randolph, B.M. Rode, *Phys. Chem. Chem. Phys.* 7 (2005) 1382–1387.
- [19] G.D. Markham, J.P. Glusker, C.W. Bock, *J. Phys. Chem. B* 106 (2002) 5118–5134.
- [20] C.W. Bock, G.D. Markham, A.K. Katz, J.P. Glusker, *Inorg. Chem.* 42 (2003) 1538–1548.
- [21] M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, J.A. Montgomery Jr., T. Vreven, K.N. Kudin, J.C. Burant, J.M. Millam, S.S. Iyengar, J. Tomasi, V. Barone, B. Mennucci, M. Cossi, G. Scalmani, N. Rega, G.A. Petersson, H. Nakatsuji, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, M. Klene, X. Li, J.E. Knox, H.P. Hratchian, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Gomperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, P.Y. Ayala, K. Morokuma, G.A. Voth, P. Salvador, J.J. Dannenberg, V.G. Zakrzewski, S. Dapprich, A.D. Daniels, M.C. Strain, O. Farkas, D.K. Malick, A.D. Rabuck, K. Raghavachari, J.B. Foresman, J.V. Ortiz, Q. Cui, A.G. Baboul, S. Clifford, J. Cioslowski, B.B. Stefanov, G. Liu, A. Liashenko, P. Piskorz, I. Komaromi, R.L. Martin, D.J. Fox, T. Keith, M.A. Al-Laham, C.Y. Peng, A. Nanayakkara, M. Challacombe, P.M.W. Gill, B. Johnson, W. Chen, M.W. Wong, C. Gonzalez, J.A. Pople, *Gaussian 03, Revision C.02*, Gaussian Inc., Wallingford, CT, 2004.
- [22] J. Labanowsky, J. Andelzelm, *Density Functional Methods in Chemistry*, Springer-Verlag, New York, 1991.
- [23] M. Alcani, O. Mo, M. Yañez, *Mass. Spectr. Rev.* 20 (2001) 195–245.
- [24] V. Tschinke, T. Ziegler, *Theor. Chim. Acta* 81 (1991) 65–78.
- [25] B.G. Johnson, P.M.W. Gill, J.A. Pople, *J. Chem. Phys.* 98 (1993) 5612–5626.
- [26] P.J. Stephens, F.J. Devlin, C.F. Chabalowski, M.J. Frisch, *J. Phys. Chem.* 98 (1994) 11623–11627.
- [27] A.D. Becke, *J. Chem. Phys.* 98 (1993) 5648–5652.
- [28] C. Lee, W. Yang, R.G. Parr, *Phys. Rev. B* 37 (1988) 785–789.
- [29] J.M. Mercero, J.M. Matxain, X. Lopez, D.M. York, A. Largo, L.A. Eriksson, J.M. Ugalde, *Int. J. Mass Spectr.* 240 (2005) 37–93.
- [30] W.J. Stevens, M. Krauss, H. Basch, P.G. Jasien, *Can. J. Chem.* 70 (1992) 612–630.
- [31] D.R. Garmer, N. Gresh, *J. Am. Chem. Soc.* 116 (1994) 3556–3567.
- [32] N. Gresh, W.J. Stevens, M. Krauss, *J. Comp. Chem.* 16 (1995) 843–855.
- [33] N. Gresh, D.R. Garmer, *J. Comp. Chem.* 17 (1996) 1481–1495.
- [34] J.M. Mercero, J.E. Fowler, J.M. Ugalde, *J. Phys. Chem. A* 102 (35) (1998) 7006–7012.
- [35] J.M. Mercero, J.E. Fowler, J.M. Ugalde, *J. Phys. Chem. A* 104 (2000) 7053–7060.
- [36] L. Rulised, J. Vondrasek, *J. Inorg. Biochem.* 71 (1998) 115–127.
- [37] J.M. Mercero, J.I. Mujika, J.M. Matxain, X. Lopez, J.M. Ugalde, *Chem. Phys.* 295 (2003) 175–184.
- [38] D.A. McQuarrie, *Statistical Mechanics*, Harper and Row, New York, 1976.
- [39] E. Cancas, B. Mennucci, J. Tomasi, *J. Chem. Phys.* 107 (1997) 3032–3041.
- [40] M. Cossi, V. Barone, B. Mennucci, J. Tomasi, *Chem. Phys. Lett.* 286 (1998) 253–260.
- [41] E. Cancas, B. Mennucci, J. Tomasi, *J. Chem. Phys.* 109 (1998) 260–266.

- [42] C.J. Cramer, D.G. Truhlar, Continuum solvation models, in: P.G. Mezey, O. Tapia, J. Bertrán (Eds.), *Understanding Chemical Reactivity. Solvent Effects and Chemical Reactivity*, vol. 17, Springer, Netherlands, Amsterdam, 2002, pp. 1–80.
- [43] D. Tunega, G. Haberhauer, M. Gerzabek, H. Lischka, *J. Phys. Chem. A* 104 (2000) 6824–6833.
- [44] A. Katz, J. Glusker, G. Markham, C. Bock, *J. Phys. Chem. B* 102 (1998) 6342–6350.
- [45] T. Dudev, C. Lim, *J. Am. Chem. Soc.* 128 (2006) 1553–1561.
- [46] T. Dudev, C. Lim, *J. Am. Chem. Soc.* 128 (2006) 10541–10548.
- [47] J. Falke, S. Drake, A. Hazard, O. Peersen, *Q. Rev. Biophys.* 27 (1994) 219–290.
- [48] T. Dudev, C. Lim, *J. Phys. Chem. B* 108 (2004) 4546–4557.